Language Models for Rational, Evidence-Driven Trading

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1 Introduction

Stock forecasting remains one of the most challenging problems in financial markets, having highly complex and unpredictable patterns affected by factors ranging from economic indicators to global events. Traditional algorithms and quantitative models have long sought to identify patterns and automate trading strategies, yet they often struggle to adapt to rapidly changing market conditions or incorporate unstructured data, such as news, social media sentiment, or corporate announcements. The ability of Large Language Models (LLMs) to analyze and interpret vast amounts of textual data has positioned them as powerful tools for stock market forecasting (Yu et al.). While large language models (LLMs) demonstrate remarkable perceptual and reasoning capabilities, translating vast amounts of numerical time series data into a consistently profitable trading strategy remains a highly non-trivial challenge.

Furthermore, obtaining transparent and interpretable explanations from a large language model about its forecasting decisions is a considerable challenge. An explainable model should be able to continuously retrieve live news and market data and generate strategies rather than just giving buy or sell signals. In this work, we are going to experiment with a wide range of approaches to combine numerical and textual data using a hybrid system of an LLM and a machine learning model to create an explainable and intelligent trader and stock forecaster that can provide strategies by analysing the state of the market.

2 Related Work

2.1 Fine-Tuning Large Language Models and Financial Sentiment Analysis

Forecasting stock market trends using neural networks remains a challenging task due to the complexity and dynamic nature of financial data (24; 43; 20). Recent studies have demonstrated that fine-tuning large pre-trained language models (LLMs) with financial data can significantly enhance forecasting accuracy (13; 11; 35; 23). For instance, Guo and Hauptmann (2024) improved stock return predictions by fine-tuning LLMs with aggregated token-level embeddings extracted from financial news. Integrating various financial data types, such as historical prices and market indicators, has also been shown to enhance model performance by aligning contextual textual information with quantitative financial metrics (23; 34). Additionally, Fu et al. (2024) explored fine-tuning LLMs for stock return forecasting using financial newsflow, highlighting the importance of integrating text representation and forecasting modules (3; 8). Furthermore, Xiao et al. (2025) proposed a retrieval-augmented generation framework for financial time-series forecasting, incorporating a fine-tuned 1B large language model (StockLLM) as its backbone, which achieved higher accuracy on financial forecasting tasks (15; 17).

Sentiment analysis using LLMs has become a crucial tool for understanding market sentiment and informing trading strategies (4; 16; 7). Bollen et al. (2011) assessed the effectiveness of models like GPT and BERT in classifying sentiment from Twitter data, revealing strong correlations between social media sentiment and market behavior. To better capture both the semantic content of news and temporal patterns in market data, hybrid approaches have emerged that combine LLMs with sequential models like LSTMs (38; 8). These hybrid models leverage textual information alongside price movements over time to generate more sophisticated and actionable trading signals (34).

2.2 LLM Agents in Market Simulation, Autonomous Trading, and Explainability

Beyond prediction, LLMs have been used to simulate market behavior through AI agents. StockAgent (45), for instance, introduced a multi-agent system where LLM-powered agents emulate investor behaviors in dynamic markets. Other frameworks, such as TradingAgents (36) and the models described in Guo and Hauptmann (2024), create agents fulfilling roles like market makers or value investors, enabling complex interactions—such as collaboration and competition—in synthetic market environments (19; 16). These models illustrate that LLMs can act as both predictors and active market participants.

More advanced systems, including FinMem (42) and TradingGPT (21), incorporate memory modules, role-playing, and instruction tuning to simulate intelligent traders capable of adapting over extended periods, highlighting the potential for autonomous financial reasoning and decision-making (21).

As financial AI models become more complex, providing interpretable explanations is crucial for maintaining user trust and meeting regulatory requirements (1; 40). The survey (26) reviews methods such as attention-based interpretability and model-agnostic techniques like SHAP and LIME (25; 31) that help users understand AI-driven financial decisions, promoting greater transparency and confidence. Emerging approaches integrate explainability mechanisms into LLM architectures to offer real-time, actionable insights for traders and compliance officers (27; 6).

3 Data

In this work, we first extract textual data from various public and semi-public sources such as TradingView, Twitter, Reddit, financial news outlets, and official speeches or announcements from market influencers (16; 4). When data is openly accessible via APIs or public web pages, we collect it directly; otherwise, we employ custom crawling and scraping tools to gather relevant content (28).

The textual data includes diverse market-relevant content such as political speeches (e.g., "Trump delivered a trade policy speech"), social media sentiment posts, analyst reports, and real-time discussions on cryptocurrencies and stocks. For example, all speeches from U.S. or European officials on economic policy can be clustered as one aggregated feature representing geopolitical market influence.

Next, we extract features from these texts using large language models (LLMs) (5; 16) that provide rich semantic embeddings capturing contextual meaning. Due to the high dimensionality of these embeddings, we reduce feature complexity by applying clustering algorithms that group semantically similar samples together (39). This process yields aggregated feature groups (e.g., clusters of speeches by American or European leaders), simplifying the feature space while retaining the most informative signals.

At inference time, these clustered features act as indicators reflecting the presence or absence of a feature within a given time window. If a feature is present, it is assigned a non-zero value corresponding to its frequency or intensity; otherwise, the feature value is set to zero.

In addition to textual features, we extract numerical features from market data, such as historical purchase volumes, current price movements, and trading activity indicators (32). Ultimately, by combining LLM-derived textual features with quantitative financial data, our model produces more accurate predictions of buy or sell signals for specific cryptocurrencies or stock assets.

For each sample, we apply binary labeling (0 or 1) to each feature which shows appearance of that feature in the sample.

After preparing the text-based data, we generate numerical data by calculating features based on the OHLCV (Open, High, Low, Close, Volume) of XAU-USD(Gold-USD) across multiple time frames to support comprehensive numerical analysis. These features are calculated by applying various trading strategies, such as SMA, EMA, MACD, and others, across multiple time frames to simplify and enhance analysis for the models. The raw OHLCV values are also included.

We will experiment with different approaches for feeding this data into the models. The first approach involves using all features directly. The second applies dimensionality reduction to the numerical features. The third clusters the numerical features together with the text-based features before feeding them into the model.

4 Approach

After preparing the text-based and numerical features described in the Data section, we feed all the features into both an XGBoost model and an LLM agent. These two models return a value of 0, 1, or 2 for each sample—where 0 indicates ”neutral,” 1 means ”buy,” and 2 means ”sell.”

We then use SHAP values from the XGBoost model and provide them, along with the original raw text data from which the features were derived, to the LLM. This allows the LLM to explain the decision-making process of XGBoost based on the Shapley values and the actual content of the input texts. By combining SHAP values with the original text, the explanation generated by the LLM truly reflects the reasons behind XGBoost's prediction, maintaining coherency and providing clear, practical justifications for the suggested action.

This approach ensures that domain experts can evaluate the decision rationale effectively and determine whether to trust and act upon the model's recommendations. Consequently, it enhances the interpretability and reliability of the trading signals, making the suggestions more actionable and trustworthy.

Additionally, the LLM predictor must also explain the reasoning behind its own decision.

We use LLMs in three different roles throughout this process. First one is sample labeling, second one is decision predicting and third one is explaining reason of model's decision. The specific LLMs used for each role will be determined in the future, based on the requirements of each use case.

To rigorously evaluate the explainability of our approach, we will utilize a labeled dataset and employ a suite of quantitative metrics, including but not limited to ROUGE and BLEU, to assess the fidelity of generated explanations (29; 22).

Beyond these conventional n-gram overlap metrics, which primarily capture lexical similarity between generated and reference texts, a comprehensive evaluation necessitates additional metrics that assess various dimensions of explanation quality. Human Sentiment Coherence (HSC) gauges the consistency of conveyed sentiment in explanations through expert human evaluation (10). Complementarily, Automated Sentiment Coherence (ASC) employs computational methods to quantify sentiment alignment automatically (14).

USR (Unique Sentence Ratio) measures the diversity of generated explanations by calculating the proportion of unique sentences relative to the total number of sentences, thus helping to detect redundancy and promote informativeness (18).

Factual fidelity is evaluated via Feature Matching Rate (FMR) and Feature Coverage Rate (FCR), which quantify the extent to which explanations incorporate and cover salient features influencing model predictions (9). To ensure the generated explanations maintain diversity and avoid redundancy, metrics such as Diversity (DIV) and DISTINCT assess the variety and uniqueness of the output (12).

The METEOR metric augments lexical similarity evaluations by considering synonymy and word order, thereby providing a more nuanced assessment of textual congruence (2). BERTScore utilizes contextual embeddings to measure semantic similarity at a deeper representational level (44).

MAUVE evaluates the trade-off between quality and diversity in language generation, particularly useful for large-scale text generation models (30). Entailment metrics ascertain logical consistency by verifying that explanations do not contradict the input data (37). Entropy (ENTR) quantifies the uncertainty or randomness in generated explanations, with higher entropy indicating greater output variability (33). Lastly, Attribute Classification Performance (ACP) measures the precision of attributing relevant features within explanations, ensuring alignment with model reasoning (37).

The choice of these metrics is informed by the nature of the explanatory outputs, the availability and scale of annotated data, and the intended application context. Employing a multi-faceted evaluation strategy facilitates the generation of explanations that are not only accurate but also coherent, diverse, and interpretable by end-users.

In addition to focusing on explainability, we aim to maximize the profitability of the model predictions. Profitability is defined as the total net profit or loss accumulated from all trades made by the models. Next, we compare the outputs of both models and investigate how the quality of explainability metrics correlates with overall profitability. We can also evaluate the quality of the extracted features and the LLM-labeled samples based on the performance of the models.

5 Schedule

We plan to work collaboratively on all components of the project. Below is a tentative timeline and task breakdown:

1. **Text-Based Data Collection and Preprocessing (5 days)**
   Crawl, scrape, and clean text data from TradingView, Twitter, Reddit, and other sources. Label the samples using an LLM.

2. **Numerical Data Collection and Feature Extraction (2 days)**
   Gather OHLCV data for XAU-USD from multiple timeframes, extract statistical and engineered features, and preprocess raw data.

3. **Dimensionality Reduction and Clustering (1 days)**
   Apply clustering techniques and reduce feature dimensionality to improve model generalization.

4. **Model Training (5 days)**
   Train XGBoost and LLM-based predictors on the combined text and numerical feature sets.

5. **Profitability and Feature Validation (3 days)**
   Evaluate the contribution of extracted features and generated labels to model profitability using XGBoost and also evaluating predictor LLM.

6. **Explainability Analysis (5 days)**
   Use Shapley values to explain XGBoost decisions. Use LLMs to generate natural language explanations for both model predictions.

7. **Evaluation and Error Analysis (3 days)**
   Perform in-depth error and performance analysis to identify weaknesses in prediction or labeling. Evaluate explainability effectiveness using human-in-the-loop assessment or proxy metrics.

8. **Final Report and Presentation (1 week)**
   Document the full pipeline, results and visualizations in the final report. Prepare the final presentation.

6 Tools

APIs

In addition to the libraries and toolkits, we will utilize several APIs to enhance the functionality and performance of our pipeline:

- **Google APIs**: These may include Google Cloud services such as the Natural Language API for advanced text analysis, Google Compute Engine for scalable computing resources, or Google Drive API for secure data storage and retrieval.

- **Deepseek API**: This API will be used. The advantage of this API is that it is really cheap while being a powerful model.

Libraries and Toolkits

To implement the described pipeline, we will utilize the following established libraries and toolkits:

- **XGBoost**: For training gradient boosting models on numerical and categorical features, with built-in support for computing SHAP values efficiently.

- **SHAP (SHapley Additive exPlanations)**: To extract and analyze feature attributions from the XGBoost model, providing consistent and interpretable explanations.

- **Hugging Face Transformers**: For integrating, fine-tuning, and deploying large language models that serve as agents for sample labeling, decision prediction, and explanation generation.

- **LangChain**: A powerful framework that simplifies building applications with large language models by managing prompt construction, chaining multiple LLM calls, and integrating external data sources. In this project, LangChain will enable us to create complex workflows where SHAP values and raw text data are combined.

- **Natural Language Processing libraries (NLTK, or Transformers tokenizers)**: To preprocess raw text data, including tokenization, normalization, and embedding tasks for effective LLM input.

- **PyTorch** : As the deep learning backends for running and fine-tuning large language models, depending on the selected architecture.

Computational Resources

For fine-tuning large language models (LLMs), we will use Google Colab's free-tier hardware, which includes:

- **GPUs:** NVIDIA T4
- **TPUs:** TPU v2-8

If additional resources are needed, we will also utilize Kaggle's free-tier accelerators, which offer:

- **GPUs:** NVIDIA T4 and NVIDIA P100

For training traditional machine learning models such as XGBoost, we will use both Colab and Kaggle CPUs when suitable.

CPU Specifications

- **Colab CPUs:**
  - Typically 2 to 4-core Intel Xeon processors
  - RAM: approximately 12.6 GB
- **Kaggle CPUs:**
  - 2 vCPUs (Intel Xeon-based)
  - RAM: 13 GB

We might rent some GPU servers for the final tuning if needed.

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